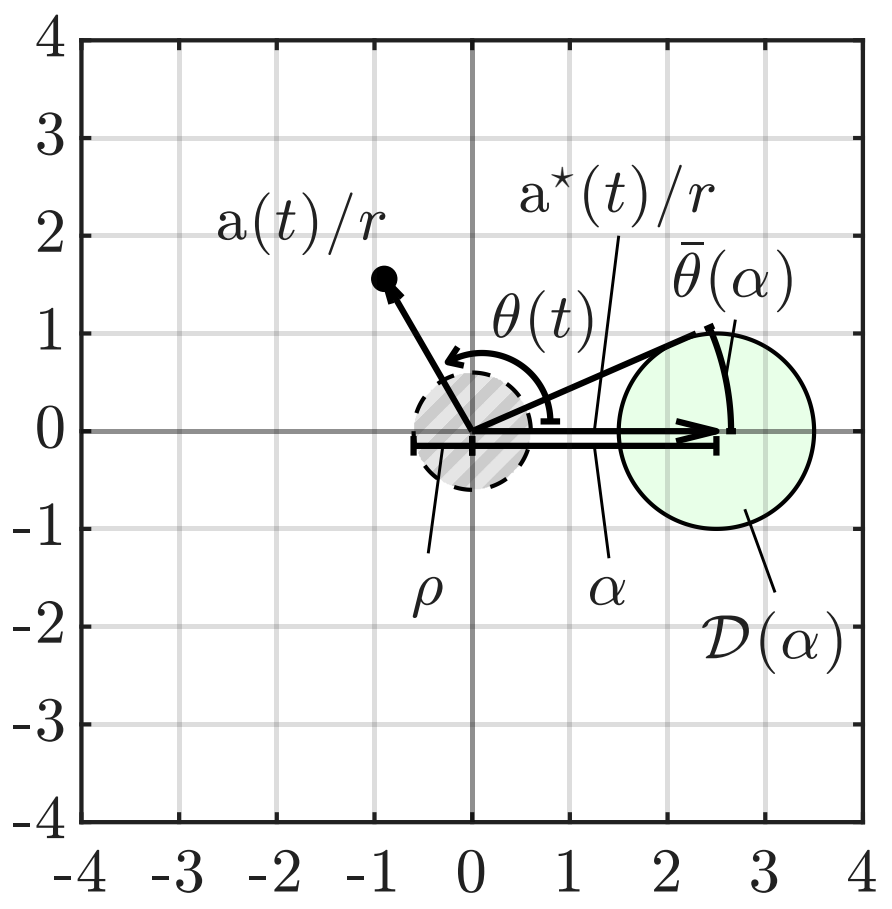

4
3
2
1
0
-1
-2
-3
-4
-4 -3 -2 -1 0 1 2 3 4
$\mathrm{a}(t)/r$
$\mathrm{a}^\star(t)/r$
$\bar{\theta}(\alpha)$
$\theta(t)$
ρ
α
$\mathcal{D}(\alpha)$